MIL-D-51248A(MU)
31 July 1968
SUPERSEDING
MIL-D-51248(MU)
18 March 1966

MILITARY SPECIFICATION

DYE, CHEMICAL AGENT INDICATOR

1. SCOPE

1.1 This specification covers one grade of chemical agent indicator dye (p-nitrophenylazo-beta-naphthylamine).

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of this issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieve, Test.

MILITARY

MIL-I-285 - Impregnite, Unstabilized (CC2) and Stabilized (XXCC3).

STANDARDS

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6820

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Physical and chemical requirements. The material shall conform to the requirements of table 1 when tested as specified therein.

Table 1. Physical and chemical requirements

Property	Requirements		Test method
	Minimum	Maximum	
Purity %	90	---	4.4.4.1
Melting point, °C	175	---	4.4.4.2
Volatile matter, %	---	0.70	4.4.4.3
Particle size, percent through No. 40 sieve	100	---	4.4.4.4
Free beta-naphthylamine,%	---	0.01	4.4.4.5

3.2 Agent indication. The material when tested as specified in 4.3.3 shall show a visible color change from a dark spot to a red or red-brown color when subjected to HD, GF, and VX. The material shall show no color change with water, Stoddard Solvent or mineral oil.

3.3 Preproduction. Prior to the start of regular production, a preproduction lot of dye shall be produced in accordance with this specification for examination and test (see 4.2).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows.

(a) Preproduction inspection (see 4.3).

(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. Prior to the start of regular production, a preproduction sample of 100 pounds minimum shall be produced using the same material and process and under essentially the same manufacturing conditions as will be used in regular production. Any change of materials or processes shall require a new preproduction sample. Approval for such change will be obtained from the Government prior to instituting the change.

4.3.2 For examination and destructive test. Two 1/2-pound specimens shall be selected from the preproduction sample and tested in accordance with 4.3.3 and 4.4.4.

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4.3.3 Tests.

4.3.3.1 Agent indication.

(a) Preparation.

(1) Dye mix. Mix thoroughly 2.5 ± 0.1 percent by weight of the dye with XXCC3 powder conforming to MIL-I-285, type II except the available chlorine may be a minimum of 10.0 percent. Allow the mixture to stand for 16 ± 1/2 hours at 25° ± 1°C and 50 ± 5 percent relative humidity.

(2) Cloth specimens. Prepare 3 individual specimens of 8-1/2 ounce, corded, cotton sateen, dyed color OD-107. Contaminate the specimens with individual spots (1 per sq cm) of HD, GF, and VX using an agent weight of 0.2 ± 0.02 mg. Allow the specimens to stand for 10 ± 1 minutes.

(b) Procedure. Dust the dye mix on the contaminated spots on each specimen and observe for the specified color change.

(c) Blank. Repeat the above procedure using 5 mg quantities of mineral oil, Stoddard Solvent and water having a pH of 4 and 10 in place of the specified agents. Observe for absence of color change.

4.3.4 Acceptance/rejection criteria. The preproduction sample shall comply with all the requirements of this specification to be acceptable.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the dye (p-nitrophanylazo-beta-naphthylamine) which has been produced by one manufacturer at one plant under essentially the same manufacturing conditions and with no change in materials provided the operation is continuous. In the event that the process is a batch operation, each batch shall constitute a lot. (See 6.3).

4.4.2 Sampling for test. Containers shall be taken at random from each lot as follows:

Lot size (in containers)	Number of containers
1	1
2 - 275	2
276 - 545	3
546 - 900	4
901 -1345	5
1346 -1875	6
1876 -2500	7

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For lots exceeding 2500 containers, the number required shall be calculated using the following equation:

$$n = 0.15\sqrt{N}$$

where:

n = number of containers
N = lot size

A 1/4 pound sample shall be removed from each container selected and placed in a clean, dry container labeled to identify the lot and the container from which it was taken.

4.4.3 Inspection for test. Each sample shall be tested as specified in 4.4.4 If any sample fails to meet the test requirements, the lot represented shall be rejected.

4.4.4 Tests. Distilled water and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.4.4.1 Purity. Determine the purity by comparing the maximum absorbance at 495 millimicrons (mμ) of a dilute solution of the dye dissolved in toluene to that of a standard sample (see 6.5) in a suitable spectrophotometer. To effect complete solution of the dye, heat dye in toluene on a steam bath until completely dissolved, then cool solution to room temperature and dilute to mark. Starting with 0.1000 grams (g) of the standard dye, prepare by successive volumetric dilution a solution having an absorbance of 0.8. Repeat the identical dilution scheme with the unknown sample and measure the absorbance at 495mμ. Calculate the purity as follows.

$$\text{Percent purity} = \frac{\text{Absorbance of unknown}}{\text{Absorbance of standard}} \times 100$$

4.4.4.2 Melting point. Grind a small portion of the sample into a fine uniform powder. Introduce a portion of the ground sample into the open end of melting point tube (available as a commercial product or formed by sealing one end of a thin walled capillary tube whose approximate length is 9 centimeters (cm) and outside diameter is 1.5 to 2.0 millimeters (mm)). Tap the sealed end of the tube gently on a solid surface to pack the sample loosely in the sealed end of the tube to a height of 4 to 6 mm. Attach the tube to a calibrated mercury thermometer, graduated in intervals of no more than 0.5°C in such a manner that the sample is adjacent to the midpoint of the thermometer bulb. Suspend

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the thermometer and melting point tube assembly in a circulating, clear, nonvolatile oil bath which has been preheated to 170°C. Temperature shall be increased at a rate of 1°C per minute. The temperature at which the sample begins to melt shall be recorded. When necessary, a correction shall be applied for the emergent stem of the thermometer.

4.4.4.3 Volatile matter. Weigh to the nearest milligram (mg), approximately 5 grams (g) of the sample in a tared, glass-stoppered weighing bottle approximately 50 (mm) in diameter by 30 mm high. Dry for four hours in an oven maintained at 115° ± 5°C, cool in a desiccator, and weigh.

$$\text{Percent volatile matter} = \frac{(\text{Loss in weight}) \ 100}{\text{Weight of sample}}$$

4.4.4.4 Particle size. Use a sieve conforming to RR-S-366 for particle size determination. Weigh to the nearest centigram (cg) approximately 10 g of the sample and place it in the No. 40 sieve. Brush the material with a camel's-hair brush until no more material passes through the sieve. Remove the sieve and weigh the material retained. Calculate the percentage of material passing through the sieve.

Note. In case of dyes too fluffy to sieve dry, the following method for the determination of particle size may be used: Weigh to the nearest cg approximately 10 g of the material and mix in a beaker by stirring with a glass rod, with sufficient water to produce a smooth paste. Carefully wash the thoroughly wetted specimen through the sieve. Dry the sieve in an oven at 115° ± 5°C and weigh the material remaining on the sieve. Calculate the percentage of material passing through the sieve.

4.4.4.5 Determination free beta-naphthylamine.

(a) Apparatus and reagents.

(1) Filter funnels - sintered glass, fine.
(2) Klett-Summerson or other colorimeter equipped with #52 (525 millimicron) filter.
(3) Hydrochloric acid solution - 0.1% volume/volume aqueous.
(4) Naphthanil Diazo Blue -B (NDBB) - 0.1% weigh/volume aqueous solution. Prepare just before use.
(5) Buffer solution: pH4.0 - add 2.0 ml of 0.1N aqueous sodium hydroxide solution to 250.0 ml 0.1M potassium acid phthalate solution. Mix and bring to 500.0 ml volume with distilled water.
(6) Beta-naphthylamine - purified and recrystallised.
(7) Ethanol - absolute.

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(b) Preparation of standard calibration curve.

Weigh approximately 0.1 gram (g) beta-naphthylamine into a 50 ml volumetric flask. Add 25 ml hydrochloric acid solution and warm the mixture until bubbles start to rise. Stopper the flask and shake mechanically while the solution cools to room temperature. Bring the flask to volume with hydrochloric acid solution. From this solution make at least four dilutions with further addition of hydrochloric acid solution until the concentration range of beta-naphthylamine is from 5 to 25 micrograms (µg) per 2.0 ml. In duplicate with corresponding blanks of hydrochloric acid solution, pipet 2.0 ml aliquots into Klett or other colorimeter tubes. Add 3.0 ml buffer solution and 1.0 ml of absolute ethanol and stir. Zero colorimeter on distilled water. Measure and record the absorbance of each solution using No. 52 (525 mµ) filter. Add 1.0 ml of MDBB reagent to each solution just measured. Stir. Measure and record second absorbance measurement of each solution at exactly 5 minutes after addition of MDBB. Correct the first absorbance measurement for volume by multyplying each by 0.857. Deduct this correction from the corresponding second measurement. Subtract the corrected average blank from the average sample measurement to obtain the net absorbance for each concentration of beta-naphthylamine. Prepare a calibration curve from these values by graphing the known concentrations versus the found absorbance and drawing a straight line through the origin and the plotted points.

(c) Procedure.

Weigh 0.1g of the dye (para-nitrophenylaso-beta-naphthylamine) into a 100 ml volumetric flask and add 50 ml of hydrochloric acid solution. Heat the solution until bubbles start to rise. Stopper the flask and shake mechanically while the solution cools to room temperature. Filter the solution through sintered glass filter funnels applying a low vacuum. Pipet duplicate 2.0 ml aliquots of the clear solution into Klett tubes and proceed as in (a) for the calibration curve procedure. Use the calibration graph to interpret the final net absorbance of the unknown in terms of beta-naphthylamine concentration. Convert the found concentration (c) of beta-naphthylamine into grams by multiplying by $50/2 \times 10^{-6}$. Divide the figure by 0.1g and multiply by 100 to obtain the percent by weight of beta-naphthylamine; i.e.

$$\frac{(c) \times 25 \times 10^{-6} \times 10^{2}}{0.1} = \text{percent by weight of beta-naphthylamine}$$

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5. PREPARATION FOR DELIVERY

5.1 Packing, level C. The dye shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall comply with Uniform Freight Classification rules or regulations of other carriers applicable to the mode of transportation.

5.2 Marking. In addition to any special marking required by the contract or order, all shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The chemical agent indicator dye (p-nitrophenylazo-beta-naphthylamine), covered by this specification, is intended for use with the M13 kit.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Preproduction

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facilities and shipping instructions when testing is performed by the Government.
(3) Time allowed for the Government to notify the supplier whether or not to proceed with production.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or a calculated value shall be rounded off "to the nearest unit" in the last righthand place of figures used in expressing the limiting value, in accordance with the rounding-off method of the Recommended Practices for Designating Significant Places in Specified Limiting Values (ASTM E29).

6.5 Standard sample. The standard sample required for determining purity in accordance with 4.4.4.1 shall be obtained from: Quality Assurance Directorate, ATTN: Chemical Laboratory Branch, US Army Edgewood Arsenal, Edgewood Arsenal, Maryland 21010.

Custodian:

Army-MU

Preparing activity:

Army-MU(EA)

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